

Should researchers create and research mirror life?

Mirror life is life based on molecules with the opposite chirality to those favoured by life because of evolution. Usually, proteins use left-handed amino acids and DNA use right-handed sugars. The engineering of mirror molecules could have prominent benefits such as better drugs, therapies that avoid evoking the immune system as well as bioreactors resistant to attack from bacteriophages and predators. However, along with this catalogue of potential rewards comes a list of risks that some researchers are already issuing warnings against. To what extent we should heed these calls for caution is to be debated below.

We shall begin with the creation of mirror molecules: proteins, DNA, RNA, ribosomes and so on. It is known by researchers that biological processes have a degrading effect on drugs. E.g. gut microbes can break down compounds like digoxin (used to treat heart failure), reducing its efficiency. These mechanisms may be specific to the chirality of the drug compounds; hence they may not affect compounds of the opposite chirality. It follows that engineering drugs with mirror molecules could greatly boost the function of a wide range of drugs, improving human wellbeing in general.

In the human body, most proteins are made of L-amino acids, so the immune system has evolved to recognise and deal with this form of chirality. This poses a problem against many forms of therapy used in medicine that end up invoking immune responses from the patient resulting in a reduction of the function of the therapy. The advent of mirror molecules could lead the creation of organs made of mirror cells that can be accepted by patients without the risk of rejection by the host immune system.

Furthermore, bacteria have formed an important part of biotechnology, a role that can be supported by mirror molecules. For example, certain bacteria can be used in conjunction with photodiodes to convert harmful greenhouse gases like carbon dioxide into other useful products. Such devices are affected by bacteriophages and predators like amoeba that kill bacteria, reducing the efficacy of the technology. Since predators have evolved to attack bacteria that contain molecules with a certain chirality, mirror bacteria, i.e. bacteria built on molecules with the opposite chirality to that found in natural bacteria, could easily be resistant to the threat of external predators and maintain their function within biotech for a longer timer.

However, the factors of success of mirror life also contribute to the dangers it poses. If bacteria gain resistance against biological mechanisms, they could easily attack and kill cells in the body unhindered. Such bacteria would easily reproduce and spread in

ecological systems where they'd outcompete other organisms for nutrients, with the potential to usurp biodiversity and wipe out whole species of organisms. It is debatable whether mirror life could make it outside the lab in the first place, let alone survive off nutrients that are suited to normal life. It is certain that mirror life could only be made in the lab and not nature. There is no reason for evolution to backtrack on billions of years of natural selection now. However, it is not uncommon for lab leaks to occur; mirror bacteria also may escape into the environment while performing the tasks it is engineered to do. I'd argue that mirror organisms would be able to survive outside the lab: such capabilities of resistance against hostile processes and predators are not likely to go unmissed. Mirror bacteria can easily develop ways to live off normal nutrients that normal organisms live off. Bacteria don't have to be oppositely chiral to the dot to still present the dangers discussed above. Some researchers claim that these fears are inflated and that many important biological processes in the body aren't specific to chirality, and it would not matter if the molecules present were mirrored or not. In contrast, the horn of bioterrorism is blown by critics, arguing that the consequences of malicious actors gaining access to such biotechnology may prove disastrous. A picture of mirror molecules specifically engineered to attack biological processes within the body that have elements of specificity to chirality is painted and provides weight to these arguments.

Many researchers argue that safety mechanisms can be put in place to make the use of mirror molecules safe. For example, engineering mirror life that has a dependence on molecules not found in nature would ensure that a mirror bacterium would not survive outside the laboratory because the molecules it needs to survive are missing. But this doesn't eliminate the malicious use of mirror life where mirror bacteria is engineered to not have a dependence on any molecules. We could also make mirror bacteriophages or antibiotics that can attack any stray mirror bacteria, killing them on command. However, it is questionable whether the rate at which such objects can be made can keep up with the rate of proliferation of mirrored bacteria. Then there is the possibility of engineering crops to be resistant to mirror life so that they could never be attacked and overrun by hostile bacteria. But it seems ludicrous that we may have to modify all the crops on earth to deal with this issue.

In conclusion, we have expressed the potential benefits of life based on oppositely chiral molecules and the dangers that such power may bring. While some criticisms against this advent of research are certainly valid, the weight that we attribute to them is not settled and it is not obvious that these concerns press us to cease research in this field and hence prevent the potential benefits that this biotechnology may bring (of which some are sure to exist). The stance that I support is to continue research in the development of mirror molecules such as proteins and DNA and to reap the benefits of

using them in the development of drugs and therapies as because the employment of these molecules is quite specific, the risk of mere molecules running havoc amongst unintended biological processes is low. However, the threats that more complex mirror organisms like bacterium and whole mirror cells pose is greater and the potential havoc that these self-reproducing entities can cause supports the argument that more research needs to be conducted into the functioning of mirror organisms and their relationship to normal nutrients in the environment before they start to be developed for commercial use. Obviously, the debate is only beginning and will continue for many more years: it is important that we heed new evidence and change our stances on the matter accordingly.